

Problem Set 1: Due on Tuesday, January 27, @noon**Dynamic Programming**

(100 points total)

1. **Reaching the center (25 points).** Consider the following static problem: an agent is located at some point x_0 on the segment $[-1, 1]$. The agent wants to reach point 0, but traveling is costly: traveling a distance d costs d^2 . After traveling, the agent pays cost D^2 if the distance from his location to point 0 is D . Therefore, the agent's problem is to choose location $x \in [-1, 1]$ that minimizes $d^2 + D^2$, with $d = |x - x_0|$ and $D = |x|$.

(a) Solve for the optimal location x .

Suppose now that the problem is repeated for T periods. (In each period, the agent pays the cost of traveling, as well as the cost according to his distance from point 0 in that period.) The agent discounts time with β .

- (b) Using induction, show that the value to the agent with n periods left takes the form $v_n(x) = A_n x^2$ for some constant $A_n \geq 0$.
- (c) Now suppose that there are infinitely many periods. Write down the Bellman equation for the agent's problem. Guess and verify (I guessed, you verify!) that the agent's value function takes the form $v(x) = Ax^2$. Notice that we can restrict our attention to $A \geq 0$, since the objective function is nonnegative. Prove that there exists a unique solution for $A \geq 0$. Give an expression for how the optimal location evolves over time.

2. **A tree-cutting problem (25 points).** A tree grows according to the law of motion $x_{t+1} = h(x_t)$, where x_t is the size of the tree at date t and h is a continuous, non-decreasing function. At date 0 you have a tree of size x_0 and you have to choose when to cut it. When you cut it, you will sell it and make a revenue of px_t , where p is the price of wood and it is constant over time. For simplicity, normalize p to one. You want to choose the cutting time T to maximize the present value of your discounted payoff, which is $\beta^T x_T$.

Make the following assumption: there is an $\bar{x} > 0$ such that $\beta h(x)/x < 1$ for all $x > \bar{x}$.

- (a) Set up the dynamic optimization problem recursively. That is, define the state variable for this problem and write the Bellman equation.
- (b) Let A be the space of continuous non-decreasing functions v that satisfy the property $v(x) = x$ for $x \geq \bar{x}$. Let T be the operator corresponding to the Bellman equation. Prove that T maps A into A .

- (c) Argue that T is a contraction in a complete normed vector space so that there is a unique solution to the Bellman equation in A .
- (d) Assume that there is an x^* such that $\beta h(x)/x > 1$ if $x < x^*$ and $\beta h(x)/x < 1$ if $x > x^*$. Prove that the optimal policy is to cut the tree as soon as it is bigger than x^* .

3. **Optimal growth model with leisure (20 points).** Consider the neoclassical growth model in which the representative household values leisure. The per period utility function is denoted by $U(c, 1 - l)$ and the production function is given by $f(k, l)$. The social planner maximizes the present discounted value of the household's utility subject to feasibility constraints.

- (a) Carefully describe how to map this problem into the Bellman equation of the form

$$v(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}.$$

That is, write down what each of the primitives X , F , Γ , and β correspond to. (Be very careful and precise!) Find conditions on the primitives of the growth model so that the function F is well defined, continuous, and increasing in its first argument. Feel free to assume that the utility function U is twice continuously differentiable.

- (b) Find a set of sufficient conditions on the primitives of the growth model that guarantee that the value function above is strictly increasing and strictly concave. (Prove that they are sufficient.)

4. **Stochastic optimal growth model (30 points total).** Consider a stochastic version of the optimal growth model. The representative agent has a per-period utility function given by $U(c)$, where c is consumption and the function U is C^2 , strictly increasing, and strictly concave. The agent is endowed with one unit of time in each period, does not value leisure, and discounts the future with $\beta \in (0, 1)$. Output in each period is produced from capital and labor according to the stochastic production function $zF(k, n) = zk^\alpha n^{1-\alpha}$, where $\alpha \in (0, 1)$. The productivity shock z has support $Z = [\underline{z}, \bar{z}]$, where $0 < \underline{z} < \bar{z} < +\infty$, with $\mathcal{Z}$ denoting the corresponding Borel sets. The shock z is i.i.d. over time, drawn according to the probability measure λ on $(Z, \mathcal{Z})$. For simplicity, assume that capital fully depreciates in each period.

- (a) Carefully write down the Bellman equation (functional equation) for this problem.
- (b) In order to use tools of bounded dynamic programming, you need to restrict the space of state variables. Determine such a constant $\bar{k}$ for which you can restrict the capital stock to lie in the interval $K = [0, \bar{k}]$.
- (c) Show that there exists a unique bounded continuous function $v: K \times Z \rightarrow \mathbb{R}$ satisfying the Bellman equation.

- (d) Show that $v(k, z)$ is strictly increasing and strictly concave in k and that the corresponding optimal policy correspondence for the next period's capital, $G(k, z)$, is a continuous single-valued function; call this function g .
- (e) Show that g has the form $g(k, z) = h(zk^\alpha)$, where h is continuous and strictly increasing. Notice that this implies that g is strictly increasing in both of its arguments. Show that the optimal consumption policy $c(k, z) = zk^\alpha - g(k, z)$ is also strictly increasing in both of its arguments.